

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET D — ASSERTION & REASONING SPECIAL

CHAPTER 05: EXPLORING MIXTURES: SOLUTION, COLLOID & SUSPENSION
TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-D-05

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET D - ASSERTION & REASONING SPECIAL):

- The question paper comprises **25 Assertion-Reasoning Questions** of **1 mark each**.
- Each question consists of two statements — **Assertion (A)** and **Reason (R)**.
- Select the correct option from the following 4 standardized choices:
 - (A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 - (B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 - (C)** Assertion (A) is true but Reason (R) is false.
 - (D)** Assertion (A) is false but Reason (R) is true.
- Section A (Q1 – Q10):** Core Theoretical Principles & Scientific Definitions.
- Section B (Q11 – Q18):** Experimental Observations & Phenomenological Inferences.
- Section C (Q19 – Q25):** Advanced Conceptual Nuances & Common Scientific Misconceptions.
- All questions are compulsory. There is no negative marking.
- Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

STANDARD DIRECTIONS FOR ALL 25 ASSERTION & REASONING QUESTIONS:

- (A)** Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

SECTION A: CORE THEORETICAL PRINCIPLES (Q.1 TO Q.10)

[10 MARKS]

Q.1

Assertion (A): A true solution of sodium chloride in water is completely homogeneous and clear.

Reason (R): In a true solution, solute particles are broken down to individual ions or molecules having diameter less than 1 nm (10^{-9} m).

- (A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. **(D)** Assertion (A) is false but Reason (R) is true.

Q.2

Assertion (A): A beam of light passed through a true copper sulfate solution does not show a visible illuminated path.

Reason (R): Particles in a true solution are too small ($< 1\text{ }\mu\text{m}$) to scatter visible light rays (no Tyndall effect).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.3

Assertion (A): Milk is classified as a colloid, specifically an emulsion.

Reason (R): In milk, tiny liquid fat droplets are dispersed uniformly throughout a continuous liquid aqueous medium.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.4

Assertion (A): A suspension is a heterogeneous mixture that is thermodynamically unstable over time.

Reason (R): The dispersed solute particles in a suspension are larger than 100 nm and gradually settle down under gravity upon standing.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.5

Assertion (A): Brass is considered a homogeneous mixture (solid solution) rather than a chemical compound.

Reason (R): Brass retains the individual chemical properties of copper and zinc, and its composition can vary within certain percentage ranges.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.6

Assertion (A): The solubility of most solid solutes like potassium nitrate (KNO_3) in water increases with an increase in temperature.

Reason (R): The dissolution of potassium nitrate in water is an endothermic process; according to Le Chatelier's principle, adding heat favors dissolution.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.7

Assertion (A): A saturated solution becomes supersaturated upon heating to a higher temperature without adding more solute.

Reason (R): Solubility increases with temperature, so a saturated solution becomes unsaturated when heated.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.8

Assertion (A): Burning of a candle involves both physical and chemical changes simultaneously.

Reason (R): Melting of candle wax and vaporizing are physical changes, while combustion of wax vapor into carbon dioxide and water vapor is a chemical change.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.9

Assertion (A): When iron filings and sulfur powder are heated strongly together, a compound iron sulfide (FeS) is formed.

Reason (R): The product FeS can be separated back into iron and sulfur using a simple bar magnet.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.10

Assertion (A): Colloidal particles do not settle down at the bottom of the container when left undisturbed.

Reason (R): Colloidal particles carry similar electrical charges that repel each other and undergo continuous random zigzag Brownian motion.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

SECTION B: EXPERIMENTAL OBSERVATIONS & INFERENCES (Q.11 TO Q.18)**[8 MARKS]**

Q.11

Assertion (A): When sunlight enters a dark dusty room through a narrow chink, the path of the beam becomes brilliantly visible.

Reason (R): Suspended airborne dust and smoke particles scatter visible light rays in all directions, demonstrating the Tyndall effect.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.12

Assertion (A): Blood cannot be separated into its cellular components by simple ordinary paper filtration.

Reason (R): Blood cells and plasma proteins are too small and flexible, easily passing through ordinary filter paper pores along with water.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.13

Assertion (A): Centrifugation is used in diagnostic laboratories to separate blood cells from liquid blood plasma.

Reason (R): High-speed spinning creates intense centrifugal force that drives denser cellular components to the bottom while less dense plasma stays on top.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.14

Assertion (A): Fog and mist are classified as liquid aerosols.

Reason (R): In fog, liquid water droplets are dispersed in a gaseous dispersion medium (air).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.15

Assertion (A): Rusting of iron is a purely physical change.

Reason (R): During rusting, iron reacts with atmospheric oxygen and moisture to produce hydrated ferric oxide ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$), a chemically distinct substance.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.16

Assertion (A): To prepare a 10% (by mass) aqueous solution of sodium chloride, 10 g of NaCl is dissolved in 90 g of water.

Reason (R): Mass percentage of solute is calculated as $\frac{\text{Mass of Solute}}{\text{Mass of Solution}} \times 100$, where mass of solution is $10 + 90 = 100$ g.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.17

Assertion (A): A suspension scatters a beam of light as long as its particles remain suspended in the liquid.

Reason (R): When the suspension settles completely at the bottom of the beaker, the supernatant clear liquid ceases to scatter light.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.18

Assertion (A): Pure distilled water is classified as a homogeneous mixture.

Reason (R): Pure water consists of hydrogen and oxygen chemically combined in a fixed ratio of 1:8 by mass and cannot be separated physically.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

SECTION C: ADVANCED NUANCES & MISCONCEPTIONS (Q.19 TO Q.25)

[7 MARKS]

Q.19

Assertion (A): Gases like oxygen and carbon dioxide dissolve to a measurable extent in water.

Reason (R): Dissolved oxygen in natural water bodies is essential for the aquatic respiration and survival of fish and aquatic plants.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.20

Assertion (A): Brownian motion provides direct macroscopic evidence for the kinetic nature and continuous random movement of fluid molecules.

Reason (R): Colloidal particles undergo erratic collisions with rapidly moving, invisible molecules of the dispersion medium.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.21

Assertion (A): Shaving cream is a colloid belonging to the category of foam.

Reason (R): In shaving cream, gas bubbles are dispersed within a liquid dispersion medium.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.22

Assertion (A): A mixture of kerosene and water can be separated using a separating funnel.

Reason (R): Kerosene and water are immiscible liquids that separate into two distinct layers based on their density difference.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.23

Assertion (A): During crystallization, pure crystals of copper sulfate are obtained from an impure solution.

Reason (R): Crystallization is superior to simple evaporation to dryness because evaporation can decompose heat-sensitive salts or leave charred residues.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.24

Assertion (A): Tincture of iodine is a liquid solution prepared by dissolving solid iodine in alcohol.

Reason (R): In tincture of iodine, iodine acts as the solute and ethyl alcohol acts as the solvent.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.25

Assertion (A): All mixtures exhibit sharp, fixed melting and boiling points.

Reason (R): Substances with fixed melting and boiling points are pure elements or chemical compounds.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET D)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & SCIENTIFIC CAUSAL EXPLANATIONS (SET D)

Q. No.	Correct	Scientific Rationale & Causal Analysis
Q.1	(A)	Assertion is true. Reason is true and correctly explains why true solutions show complete microscopic homogeneity.
Q.2	(A)	Assertion is true. Reason is true and provides the optical explanation for absence of Tyndall effect in true solutions.
Q.3	(A)	Assertion is true. Reason is true and defines an emulsion (liquid dispersed in liquid).
Q.4	(A)	Assertion is true. Reason is true and explains why suspensions settle out while true solutions remain stable.
Q.5	(A)	Assertion is true. Reason is true and explains why alloys are classified as mixtures rather than fixed stoichiometric compounds.
Q.6	(A)	Assertion is true. Reason is true and provides the thermodynamic explanation for temperature-dependent solubility.
Q.7	(D)	Assertion is false: Heating a saturated solution makes it unsaturated (it can dissolve more solute), not supersaturated. Reason is true.
Q.8	(A)	Assertion is true. Reason is true and distinguishes the simultaneous physical state changes from the chemical oxidation reaction.
Q.9	(C)	Assertion is true. Reason is false: FeS is a new chemical compound where iron is chemically bonded; it is non-magnetic and cannot be separated mechanically.
Q.10	(A)	Assertion is true. Reason is true and explains the kinetic and electrostatic factors keeping colloids stable.
Q.11	(A)	Assertion is true. Reason is true and explains the optical manifestation of Tyndall scattering by aerosol particles.
Q.12	(A)	Assertion is true. Reason is true and explains why centrifugation is required instead of gravity filtration.
Q.13	(A)	Assertion is true. Reason is true and describes the physical principle of centrifugal phase separation.
Q.14	(A)	Assertion is true. Reason is true and matches the IUPAC classification: liquid dispersed in gas = liquid aerosol.
Q.15	(D)	Assertion is false: Rusting is a chemical change with new bond formation and mass gain. Reason is true.
Q.16	(A)	Assertion is true. Reason is true and provides the exact mathematical definition of mass percentage concentration.
Q.17	(B)	Both statements are true experimental observations, but Reason describes what happens after settling rather than explaining why suspended particles scatter light.
Q.18	(D)	Assertion is false: Pure water is a pure chemical compound, not a mixture. Reason is true (it defines a compound with fixed mass composition).
Q.19	(B)	Both statements are factually true, but Reason describes the biological importance rather than explaining the intermolecular dipole-induced dipole forces causing gas solubility.
Q.20	(A)	Assertion is true. Reason is true and correctly describes the kinetic molecular mechanism uncovered by Robert Brown and explained by Einstein.
Q.21	(A)	Assertion is true. Reason is true and gives the standard colloidal definition of foam (gas in liquid).
Q.22	(A)	Assertion is true. Reason is true and provides the physical principle governing separating funnel extraction.
Q.23	(B)	Both statements are correct, but Reason explains why crystallization is preferred over evaporation, not how crystal lattice exclusion purifies the salt.
Q.24	(A)	Assertion is true. Reason is true and correctly identifies the solute and solvent in this pharmaceutical solution.

Q. No.	Correct	Scientific Rationale & Causal Analysis
Q.25	(D)	Assertion is false: Mixtures melt and boil over a range of temperatures, not at a fixed point. Reason is true (pure substances have sharp physical constants).